Enhancing Bagging through Signal-Noise Separation: A Novel PCA-Based Hybrid Framework for Imbalanced Classification

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Abstract

Class imbalance in classification problems can significantly impair model performance, particularly for minority classes. This paper introduces a novel hybrid approach to address class imbalance in classification problems by enhancing bagging ensemble methods through principal component analysis (PCA). The proposed framework utilizes truncated singular value decomposition to split the data into a low-rank approximation and noise components. These components are separately bootstrapped to create synthetic balanced samples that are then passed to individual classifiers. The method aims to preserve the underlying data structure while introducing variability, potentially improving robustness and generalization beyond the original dataset. Another one of our objectives is the principle of parsimony, in a sense that our method should be comparatively computationally inexpensive. The approach is applied to and evaluated on a Taiwanese bankruptcy prediction dataset collected from the Taiwan Economic Journal for the years 1999 to 2009.

Keywords: Classification, Random Forest, Class Imbalance, Bagging, Principal Component Analysis, Singular Value Decomposition, Bankruptcy Prediction

1. Introduction

A primary goal of data mining and machine learning is the construction of models that can effectively analyze and make predictions on complex datasets. The process of generating such models is called "learning" or "training" and generally falls into two categories: supervised learning and unsupervised learning Zhou (2012). In supervised learning, the setting discussed in this paper, the goal is either classification, if the target variable is categorical, or regression, if the target variable is numerical. One problem that classification tasks run into is class imbalance. For example, say our goal is financial fraud detection. Unfortunately, in practice financial fraud datasets suffer from a severe class imbalance issue, where the fraud class proportion is significantly lower than that of non-fraud. In 2022, for example, only 3% of international eCommerce orders were fraudulent, even lower than the 3.4% of orders in the previous year (Cybersource). This severe imbalance presents a challenge in traditional learning algorithms, because their goal is to maximize prediction accuracy. If the model

can achieve 97% accuracy by simply favoring the majority class, i.e. non-fraud, it becomes useless for the task at hand. Learning from data with an imbalanced class distribution, referred to as imbalanced learning, is one of the most formidable challenges within data mining.

Besides financial fraud detection, imbalanced data classification encompasses many other applications, such as medical diagnosis or bankruptcy prediction and has gained significant importance due to a growing volume of data from intricate, large-scale domains. Consequently, over the past decades, imbalanced learning has amassed broad research Chen et al. (2024) and numerous methods have been proposed to tackle it. The general methods existing in current literature can be categorized into three main approaches. [1]

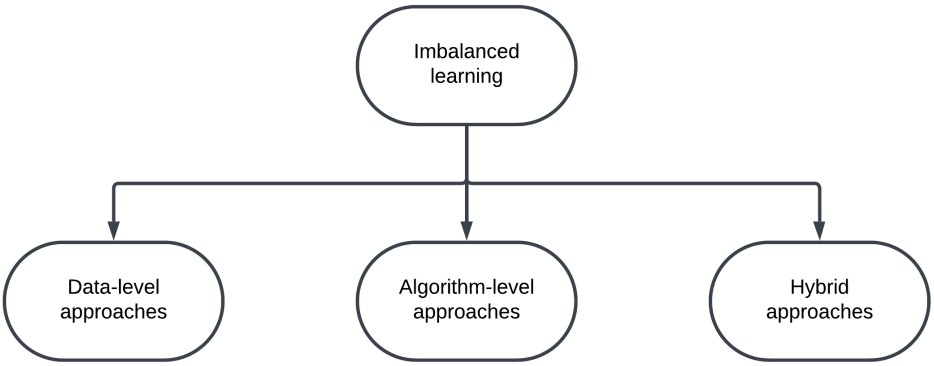


Figure 1: Imbalanced learning approaches.

1.1 Data-level approaches

Perhaps the simplest to implement, data-level approaches, address class imbalance by directly modifying the training dataset. One data-level method is to randomly undersample the majority class. In cases of mild class imbalance, this method, when applied carefully, can work. However, in cases of more severe imbalance as in the aforementioned fraud detection example, this results in too large a loss of information, which can introduce unwanted bias, particularly for the majority class. Another data-level approach to address imbalance is random oversampling of the minority class. However, random oversampling also comes with its own set of challenges. While preserving all majority class information, random oversampling can lead to overfitting, in this case for the minority class. By creating exact copies of existing minority samples, a model may become too specialized to these examples, potentially compromising its ability to generalize to new, unseen data. Given these limitations of both random undersampling and oversampling, researchers have developed more sophisticated techniques to address class imbalance. One such method is the Synthetic Minority Over-sampling Technique (SMOTE) Chawla et al. (2002), which instead of simply duplicating existing minority class instances, generates synthetic ones by applying linear interpolation between a minority class point and one of its K nearest neighbors. However, this method is not grounded in solid mathematical theory. Despite yielding good results in some applications, it has been shown to have several limitations Alkhawaldeh et al. (2023).

One glaring issue with SMOTE is that generating new minority class examples based solely on their similarity to existing ones, can falsely generate minority class examples that may actually belong to a different class in the real world. It has also been shown that instances produced by SMOTE may not precisely match the original distribution of the minority class Elreedy et al. (2023). The risk of a model failing due to training on potentially misrepresented or false data is a significant concern, especially in applications where reliability and accuracy are critical like medical diagnosis.

1.2 Hybrid approaches

A solution to many of the limitations of data-level approaches is to combine sampling methods with ensemble learning techniques. One example of an ensemble learning technique is bagging, short for bootstrap aggregating, that creates multiple training subsets by randomly sampling with replacement from the original dataset. By training separate base learners on different subsets and then aggregating their predictions with voting or averaging, bagging helps reduce variation in predictions and improves model generalization and robustness. Rather than trying to construct a single strong learner, bagging by combination of multiple learners can help find the best local optima. Although it can improve overall classification performance, the standard way of bootstrapping the original dataset may still struggle in cases of severe class imbalance. Statistically, the under-represented class has a much lower chance of being chosen in the bootstrapping process, meaning the majority of learners are likely to predict the majority class, perpetuating the bias. By itself, bagging does not address the fundamental imbalance issue. However, when combined with data-level approaches the combination can be useful. As mentioned before, random undersampling in the data-level approach leads to a loss of information, but if we create subsets of the dataset by artificially undersampling the majority class, we can effectively balance the classes and use all the information present Chawla et al. (2002). Another approach that has shown promise, is noisy replication of the minority class Lee (2000). By oversampling the minority class with a small level of normally distributed noise, we can not only decrease the imbalance of classes but also improve the robustness of the final prediction. That is, the addition of noisy replicates effectively serves as a regularization technique, solving the overfitting problem of random oversampling. However, this method also comes with some limitations. In summary, the method requires many assumptions to be made about the noise distribution characteristics. For one, the noise standard deviation σ_{noise} needs to be chosen empirically, and the method of choice proposed by the authors involves training multiple models and evaluating their results, which can become computationally expensive. Second, the method uses a diagonal covariance matrix, whereas the "true noise" may be correlated. And finally, the choice of normal distribution is also arbitrary and may not work in cases where the distribution of the "true noise" has heavy tails. Of course, these problems can be tackled, but may add additional computational complexity to a method that already suffers from it. We will attempt to overcome many of these issues present in noisy replication by taking a different approach.

1.3 PCA-Based Signal-Noise Separation

As it has been shown that adding noise to the under-represented class can work as a regularized class balancing method, our aim is to create more diverse and representative synthetic samples, potentially improving the ensemble's ability to generalize to unseen data. For the method to be an improvement, we are looking for 3 important requirements:

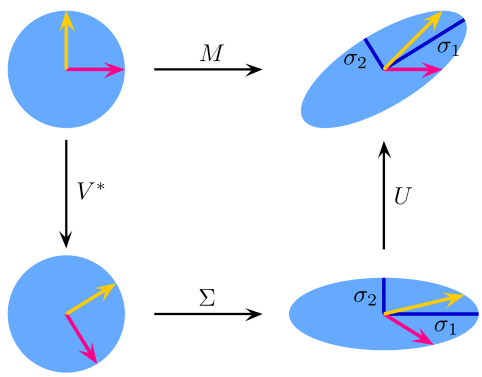
1. The added noise should have some some intrinsic characteristics that match the original data.
2. The method should make the least amount of assumptions about the noise.
3. The method should be computationally inexpensive.

That is, if possible we want to find a representation of our original data X in the form

$$X = \hat{X} + \epsilon \tag{1}$$

Where $\hat{X}$ is some "ground truth" signal and ϵ is the noise we wish to add. We can express our original data matrix through singular value decomposition [2] Kuang (2015)

$$X = U\Sigma V^T \tag{2}$$



$$M = U \cdot \Sigma \cdot V^*$$

Figure 2: Visualization of SVD. Wikipedia contributors (2024)

We will take X to be centered and normalized in this case, as that seems to work better for signal extraction Wall et al. (2003). Now, it can be shown Roughgarden and Valiant (2024) that taking the truncated U, Σ and V matrices [3], we can obtain an approximation of X

$$X_k = U_k \Sigma_k V_k^T \tag{3}$$

that is the best k-rank approximation of X in the Frobenius norm, which we will call the 'signal'. By taking

$$\epsilon = X - X_k \tag{4}$$

as the noise, we obtain precisely what we were looking for. By separately bootstrapping and adding the samples together we can create synthetic points not present in the original data. Our first two requirements are satisfied, since X_k is a structured approximation of the original data and the noise ϵ is simply taken as a difference that maintains any correlations or tail behavior. Also, this method is now model-free and efficient algorithms exist to calculate the SVD, so the third requirement is also satisfied.

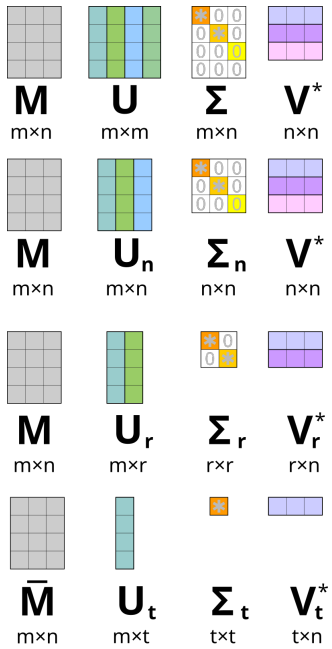


Figure 3: Illustration of truncated SVD matrix multiplications. Wikipedia contributors (2024)

As a side note, this truncated SVD approximation is exactly the same as the one that could be obtained by principal component analysis. Since PCA is now often calculated using SVD and there is effectively no difference between the two methods for centered data, we will refer to this method as 'PCA-Based', simply because the term 'principal component' captures the inherent importance of the chosen rank k much better than the term 'singular value'.

2. Data

2.1 Data set

In this paper, we utilize a dataset designed for predicting bankruptcy among Taiwanese companies, publicly available on the UCI Machine Learning Repository. The data was collected from the Taiwan Economic Journal for the years 1999 to 2009. Company bankruptcy was defined based on the business regulations of the Taiwan Stock Exchange. The dataset contains 6,819 cases and 96 variables, including our target variable that indicates the bankruptcy status of each company.

All variables are numerical, consisting of 8 integer-type and 88 continuous variables. The dataset captures a broad range of economic indicators, such as interest rates, tax rates, asset turnover, and other key financial metrics.

This same dataset was previously used in Wang and Liu (2021), where a framework was used to spot-check algorithms quickly and combine which undersampling method and classification model performed best. The undersampling methods used where: Tomek Links (TL), Edited Nearest Neighbors (ENN), Repeated Edited Nearest Neighbors (RENN), One Side Selection (OSS), and Neighbor Cleaning Rule (NCR).

2.2 Exploratory data analysis

First we can check some basic statistics. Since one of the main things we want to investigate in this paper is class imbalances, we can begin by examining the target variable to assess the extent of class imbalance in our dataset.

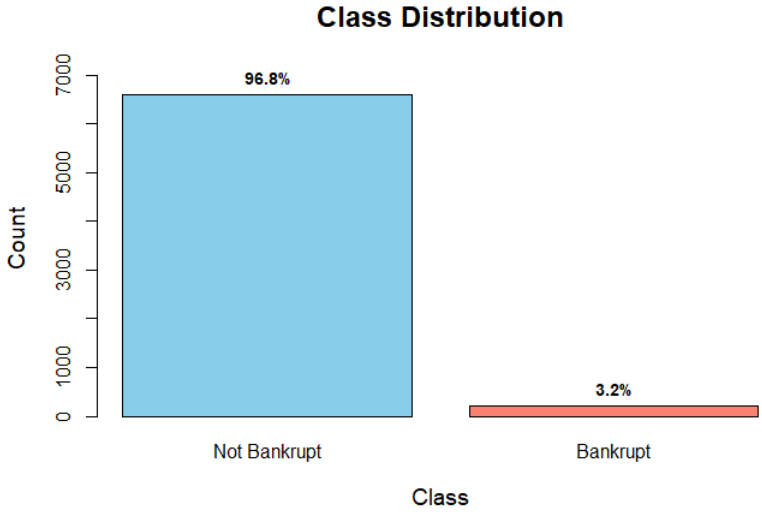
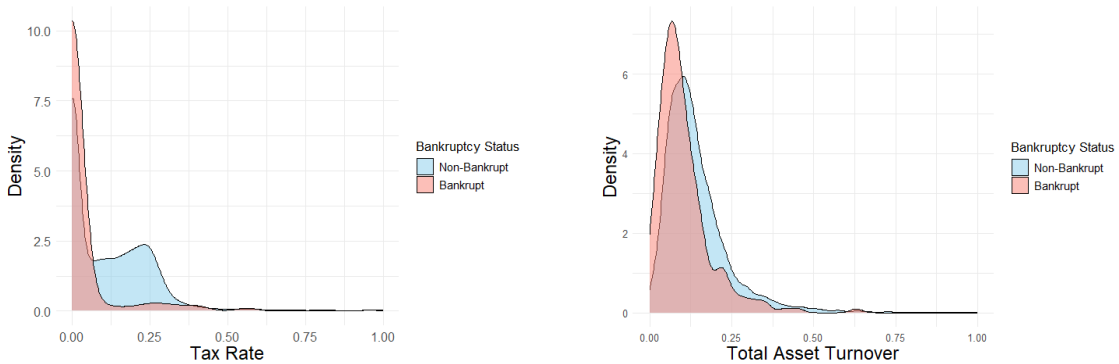


Figure 4: Class Imbalance of bankruptcy status

In figure 4, we observe a significant imbalance. Out of our 6819 companies, 6599 have not gone bankrupt, while only 220 have. In percentage terms, this translates to 96.77% of companies remaining non-bankrupt and just 3.23% going bankrupt. This is crucial to our thesis, as it is precisely the issue that we are trying to address.

As we must retain some proportion of the bankrupt classes, we will use stratified sampling to split the dataset into equal sized training and testing samples. The 50% split is unusual, but required for an adequate evaluation of model performance on the minority class.

Another thing that we can check, is the density of some of the features by bankruptcy status:



(a) Tax Rate by Bankruptcy

(b) Total Asset Turnover by Bankruptcy

Figure 5: Density Plots

In the above plots, we can see the overlapping density of Tax Rate and Total Asset Turnover by bankruptcy status. From figure 5a, we can see that for a variable like Tax Rate, the bankrupt company curve is skewed much lower, which means that a lower tax rate may be associated with bankruptcy. On the other hand, when looking at the density of a variable like Total Asset Turnover in figure 5b - the curves are overlapping a lot more so there might not be much difference between these groups. Nevertheless, the first plot of Tax Rate densities indicates a feasibility of constructing a predictive model for the target variable.

3. Methodology

The bagging methods will be applied for one underlying model - decision trees. We also do feature subsampling as is standard in random forests, using the root rule for the feature count.

Overall, we will be comparing 4 methods:

1. Standard Bagging (Rf)
2. Random Undersampling (RfRU)
3. Oversampling with Gaussian noise (RfGO)
4. PCA noise separation through truncated SVD (RfPCA)

All methods will have an ensemble of 50 trees. Also, both the Gaussian and PCA methods will have their noise levels set to 5% (based on variance) so that the comparisons

are fair. In our case, this translates to a total of 50 principal components used in the signal (based on the sum of squared singular values).

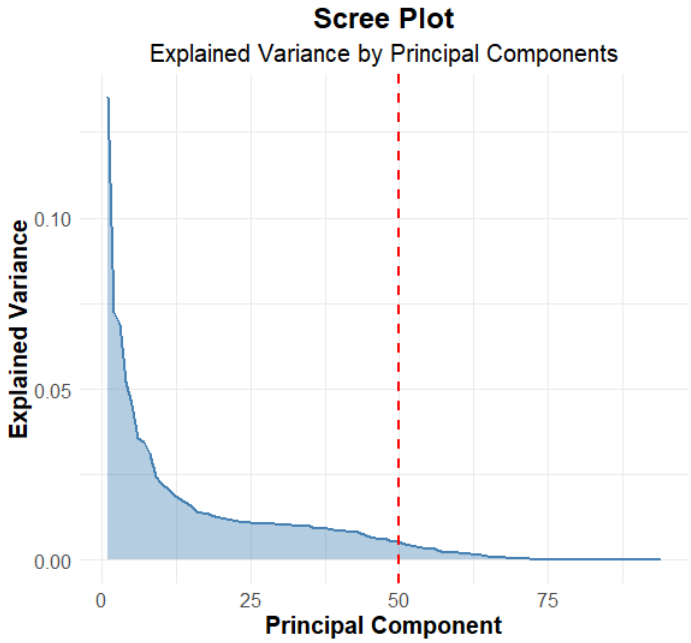


Figure 6: Scree plot of PCA.

All the methods are evaluated based on their out-of-sample sensitivity and specificity results. These show the true positive and true negative rates for our predictions, respectively. Given the significant class imbalance in our dataset, sensitivity and specificity are more appropriate metrics to evaluate model performance than the overall accuracy score. Using these metrics we can also derive a 'balanced accuracy' score as the average of the two. All of our code is publicly available on github Žurauskas E. et al. (2024).

4. Results

The results of our analysis indicate the effectiveness of our proposed method in handling imbalanced classification tasks. Tables 1 and 2 present the results of the confusion matrices and performance metrics, respectively, for our approach (RfPCA) compared to the three other approaches (Rf, RfRU and RfGO).

Unsurprisingly, the confusion matrix results in Table 1 reveal that the standard bagging method failed to detect any true positives, demonstrating its inability to handle severe class imbalance. However, what is more surprising is that oversampling with Gaussian noise (RfGO) also showed particularly poor performance. Given that the method is designed to address class imbalance, performing as poorly as the baseline Random Forest model was most unexpected. This suggests that the synthetic samples generated through Gaussian noise oversampling may not have correctly captured the distribution of bankrupt cases. Granted, this could be due to the arbitrary choice of noise level or the data normalization, but it still shows a significant drawback of the method.

In contrast, both RfRU and RfPCA show significantly better performance. RfPCA successfully identified 94 true positives followed by the 85 of RfRU, while maintaining a relatively low false negative count of 16, followed by the 25 of RfRU. These results are also reflected in the performance metrics of Table 2, showing that while RfPCA achieved a slightly lower overall accuracy (87.15%) and the highest count of false positives, its balanced accuracy was the highest of all evaluated models at 86.33%, due to the much higher sensitivity of 85.45%. In addition, we can observe that the difference between sensitivity and specificity is the smallest in the RfPCA model compared to all the evaluated models, demonstrating the most balanced performance across classes.

Model	**TP**	**FP**	**TN**	**FN**
Rf	0	0	3299	110
RfRU	85	294	3005	25
RfGO	0	1	3298	110
RfPCA	94	422	2877	16

Table 1: Confusion matrix results

Model	**Accuracy**	**Balanced Accuracy**	**Sensitivity**	**Specificity**
Rf	0.9677	0.5000	0.0000	1.0000
RfRU	0.9064	0.8418	0.7727	0.9109
RfGO	0.9674	0.4998	0.0000	0.9997
RfPCA	0.8715	0.8633	0.8545	0.8721

Table 2: Performance metrics

5. Conclusions

Our results reveal interesting findings regarding the performance of different bagging methods for imbalanced classification. Despite having a higher false positive rate, the proposed RfPCA model demonstrated superior performance on the minority class by correctly identifying 94 true positive cases. In real-world applications such as creditworthiness assessment, this makes it the most practical model to use, since in the context of bankruptcy prediction false negatives normally come at a higher cost than false positives. As this was precisely the aim of our paper, these findings are promising and suggest that our proposed method allowed the model to better generalize beyond its training data. Further analysis would require the method to be evaluated on different types of datasets and underlying model classes. We must also acknowledge that the approach is not suitable for categorical features and sparse data, at least from the theoretical side. Another surprising result was the severe underperformance of the Gaussian oversampling method. While we did acknowledge the theoretical limitations of Gaussian oversampling, this poor performance in practice was unexpected. It would be interesting to see a more in-depth investigation of the effectiveness of synthetic Gaussian noise samples in capturing different class distributions. In summary, the findings of our study show promising results and offer much room for further analysis.

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